Exploring Zoological Diversity and Ecotoxicological Impact Assessment

INTERNSHIP REPORT

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PREFACE

This report summarizes my summer internship under the guidance of Dr. Reshma Sinha at the Central University of Himachal Pradesh. The aim was to optimize DNA extraction protocols across different species with an emphasis on efficiency and reproducibility.

Model organisms included mosquitoes, earthworms, and zebrafish, through which I gained handson experience in handling, culture, nucleic acids isolation, and oxidative stress-based toxicological assessments. I also received introductory training in ELISA systems by Miss Navya.

In addition to this, I performed chemical preparation and routine physiological assessments—including blood pressure, glucose, hemoglobin, blood group, pH, and TDS measurements in complementation to the main objectives.

I appreciate the focused guidance provided by Miss Poonam Kumari in DNA and vector work, Miss Naveeta in reviewing scientific drafts and Miss Yogita in earthworm studies.

Scope of Work and Methodological Training

1. RATIONALE BEHIND ORGANISM SELECTION AND EXPERIMENTAL RELEVANCE
    - Model Organisms
    - Morphological differences in mosquitoes for species identification
    - General anatomy
2. MOLECULAR WORKFLOW OPTIMIZATION ACROSS TAXA
    - Introduction to DNA Extraction
    - Proposed Modification
    - New Protocol for extracting DNA
3. COMPARATIVE GENOMIC MATERIAL EXTRACTION AND QUALITY ASSESSMENT
    - Instrumental differences between NanoDrop & Conventional Spectrophotometer
    - Interpretation of absorbance readings and nucleic acid quantification values
    - Analysis of DNA quality using gel electrophoresis
4. FUNCTIONAL BIOMOLECULAR ANALYSIS AND TOXICODYNAMIC ANALYSIS
    - Nuclear Abnormalities and Histological Staining
    - Enzymatic Analysis
    - Hematological Indices and Dilution Factors
5. SUPPLEMENTARY EXPERIMENTAL TECHNIQUES AND ANALYTICAL EXPOSURE
    - Introduction to ELISA
    - General Procedures
    - Digital Softwares

1. Rationale Behind Organism Selection and Experimental Relevance

1. MODEL ORGANISMS

Model organisms are not just convenient species, they are research-oriented organisms selected through rigorous methods to understand biological functions across species. A short life cycle, ease of laboratory rearing, and critical role in assessing biological modifications make a species model. Besides this, the choice of a model organism depends on relevance of studies on cancer, emerging fields like nanotoxicity and brain diseases (like Alzheimer's) as seen in Drosophila Melanogaster . Our study includes Mosquitoes, predominantly followed by zebrafish, termites and earthworms.

1. Mosquitoes: Hematophagy-induced gene regulation, vector-pathogen co-evolution, and chemosensory-driven host detection are the key indicators of mosquitoes being model organisms. Their morphological differences are fascinating and include differences in color and denticles of the larva while varied strip or number of dots on its thorax.

2. Earthworms: In soil ecotoxicology, earthworms serve as bioindicators because of their oxidative stress responses, metal-binding proteins, and cutaneous respiration. Their coelomic fluid-based immunity allows functional assays of pollutant bioaccumulation. Morphologically, they display clitellum positioning, segmental setae patterns, and pigmentation varying with substrate exposure.

3. Termites: Gut-mediated lignocellulose degradation, microbial symbiosis, and anaerobic enzymatic pathways involved in detoxification makes termite an ideal emerging model organism.

4. Zebra fish: Transparent, rapidly developing vertebrate model organism widely used in genetics, toxicology, developmental biology, and disease research.

MOSQUITOES:

Mosquitoes have a diverse collection of genus and species, including medically significant taxa like Aedes, Anopheles, and Culex, which can easily be differentiated by major morphotypes like scutal patterning, siphon length, larval denticles, and thoracic striping.

Figure 1: Representative micrographs showing developmental progression through successive instars of male *Aedes albopictus* having serrated denticles on comb scales and three distinct dark thoracic dots in larva and adult mosquito, respectively; observed under dissecting microscope at 10X. a) Eggs b) i. Larva ii. Denticles c) Pupa d) Late stage in pupa e) Adult mosquito

Figure 2:Ambigous Larva 10X

Figure3:Adult culex 10X

Nonetheless, some specimens had intermediate or ambiguous traits that made it difficult to identify them definitively at the species level. These unidentifiable individuals can be intraspecific variations, locally uncharacterized species, or physically cryptic taxa that need molecular barcoding. An unidentified larva of Aedes sp. having denticles on the dorsal and ventral sides. Anal gills have a leaf-like symmetry, and anal brushes are clearly visible. But barcoding analysis is essential to characterize the species type.

Culex quinquefasciatus larvae possess a short, stout, and cylindrical siphon with a dense array of pecten denticles (ranging from 20 to 45), which are shorter, blunt, and arranged in a continuous comb-like fashion along much of the siphon's length.

The siphonal hair tufts also differ: Aedes aegypti larvae have a single pair of subapical tufts, while *Culex quinquefasciatus* displays multiple pairs (up to five), irregularly distributed along the siphon. Anal papillae in Aedes are long and slender, whereas those in Culex are shorter and blunter. The head of Aedes larvae shows fewer, longer lateral setae, while Culex larvae are characterized by a dense covering of shorter bristles, giving them a bushier appearance. On the eighth abdominal segment, Culex larvae show a prominent saddle with ventral setae, which is absent or less pronounced in Aedes.

Behaviorally, Aedes aegypti larvae typically float horizontally near the water surface, skimming clean artificial containers such as tires, pots, and tanks. In contrast, Culex quinquefasciatus larvae often hang at an angle, feeding in polluted or organically rich waters like drains and sewage pools. The siphonal denticle pattern—few serrated teeth in Aedes versus many tightly packed blunt teeth in Culex—remains the most reliable microscopic feature for differentiating these species, as consistently described in WHO larval identification manuals and entomological studies by Harbach and Knight.

Fig. 4 represent Culex spp. Having different denticle patterns.

Figure 5: a) short, thick siphon: Culex b) Aedes spp.

Figure 6: Dissected *Aedes albopictus*

Despite microscopic examination and comparison with standard larval keys, the precise identification of the species remains uncertain due to overlapping morphological traits and limited availability of definitive reference samples at the time of observation. Further validation using molecular markers or expert taxonomic input may be required.

ANATOMY OF MOSQUITO

1. Top dark part – That's probably the head of the mosquito. It has the eyes, antennae (for smell), and palps (small feeler-like parts). These help the mosquito sense

where you are — your body heat, your smell, and even the carbon dioxide when you breathe out.

2. Middle tube-like structure – That’s the proboscis, or the feeding tube. It looks like one tube, but it actually has several tiny needle-like parts inside. These work together to pierce the skin, release saliva (which prevents blood clotting), and suck blood — all at the same time.

3. Lower branching part – These look like thin veins or nerves coming down. These could be part of the salivary ducts or nerves connected to the mouthparts. Some of this could also be the inner muscles that control how the proboscis moves.

In conclusion, the mosquito's head and proboscis are highly specialized structures adapted for sensing, piercing, and feeding. Each part from the sensory antennae to the complex needle-like stylets works together efficiently, enabling the mosquito to locate its host, pierce the skin, and draw blood, all while delivering saliva that facilitates feeding. The anatomy of mosquito has played an important role in its evolution as an effective and adaptable vector.

2. Molecular Workflow Optimization Across Taxa

DNA extraction procedures need to align with distinct biochemical and structural traits, implying a clear difference in extraction protocols for organisms like termites (Isoptera) and mosquitoes (Culex and Aedes spp.). This is indeed important to ensure workflow across the molecular techniques including PCR (Polymerase Chain Reaction) and Sequencing. Without compromising yield or quality of DNA, the protocol aims to speed the two days general protocol to an optimum one-day protocol. This chapter covers overall experimental based modifications in the protocol.

Sample size: The estimation of sample size depends on the DNA content present per cell, body parts having least contamination and microorganisms. For instance, the termite head is used for DNA extraction as its gut is contaminated and 3–5 heads are appropriate for sequencing ahead [min (0.4 pg/ cell * 1M cells = 400 ng)].

Figure 7: Termite washing

The standard protocol includes dissection, homogenization in lysis buffer, incubation at 37 degrees Celsius water bath, Proteinase K → incubation at 55 degrees Celsius, PCI (Phenol Chloroform Isoamyl alcohol) step, transferring aqueous layer, precipitation with sodium acetate, 70% ethanol wash, pellet drying and DNA suspension.

Figure a: Termite head

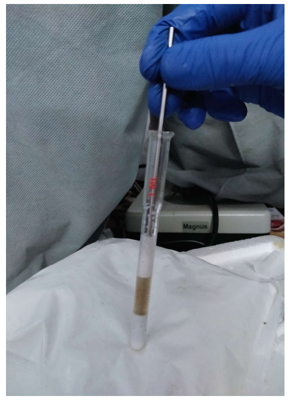


Figure b: Homogenization

Figure c: water incubation

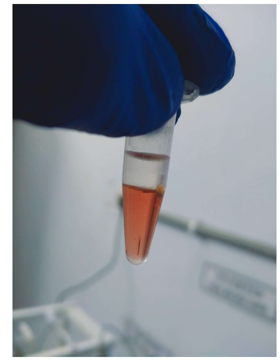


Figure d: Aqueous layer

Figure d: Aqueous layer Figure e: Dissolved in TE buffer

Figure 8: Stepwise protocol to extract DNA

Proposed Modifications:

1. Add 0.5% Tween 20 and 0.5% Triton X-100 which helps cell solubilize lipid bilayers and degrade membranes gently.

2. To further purify the sample after PCI treatment, especially when not using Proteinase K, an intermediate chloroform: isoamyl alcohol (24:1) (CI) cleanup step can be introduced after lysis and before precipitation.

3. Overnight precipitation can be avoided by using isopropanol which precipitates DNA rapidly by decreasing the dielectric constant of the solution which makes DNA less soluble and tends it to aggregate.

4. For mosquito full body extracts, where lipid and protein content are higher and microbial DNA may be present, use sodium acetate (3 M, pH 5.2) combined with cold isopropanol (0.7 volumes) for efficient precipitation of both host and microbial DNA, ensuring good yield.

5. For termite head-only samples, where cleaner, protein-limited extracts are desired, especially without gut interference, potassium acetate (3 M, pH 5.5) is preferable as it helps remove SDS-protein complexes effectively before alcohol precipitation, followed by either isopropanol or ethanol (2 volumes) for high-purity genomic DNA.

6. NFW vs TE Buffer: NFW is ideal for enzyme reactions due to its purity but lacks DNA stability protection, while TE buffer protects DNA long-term with Tris-EDTA but may inhibit enzymes due to EDTA chelation with magnesium ions.

New Protocol:

Genomic DNA was extracted from mosquito whole-body samples and termite head-only samples using a modified detergent-assisted organic extraction protocol. Tissue samples were homogenized in lysis buffer containing SDS supplemented with 0.5% Tween-20 and 0.5% Triton X-100. Although SDS is a strong anionic detergent capable of disrupting lipid bilayers and denaturing proteins, it was found to be insufficient when used alone for lipid-rich insect tissues. Mosquito whole bodies contain significant lipid content, membrane-bound organelles, and structurally complex tissues that can limit efficient solubilization by SDS alone. Moreover, excessive reliance on SDS may lead to strong protein-DNA interactions, trap DNA molecules, incomplete lipid solubilization, increased viscosity due to the presence of cellular debris, and potential DNA shearing. Therefore, Triton X-100, a milder non-ionic detergent, was included to enhance membrane solubilization while minimising excessive protein denaturation & Tween-20 for gentle lipid solubilization and improved homogenization by reducing surface tension. The combined use of these three detergents ensured efficient and balanced lysis, improved DNA release, better handling of lipid-rich material, and preservation of DNA integrity.

We did not include a 37 °C step because the protocol is chemically driven rather than enzyme dependent. The homogenate was incubated at 55–60°C to promote complete membrane disruption and protein denaturation for 35-40 minutes.

After lysis, an equal volume of phenol:chloroform:isoamyl alcohol (25:24:1) was added to denature proteins and separate them from nucleic acids. Phenol disrupts protein structure, chloroform enhances phase separation, and isoamyl alcohol reduces foaming during mixing. After centrifugation, the aqueous phase containing DNA was carefully transferred to a fresh tube. To improve purity, particularly when Proteinase K was not used, an additional chloroform:isoamyl alcohol (24:1) cleanup step was included to remove residual phenol and protein contaminants.

For mosquito whole-body extracts, which contain high lipid and protein content and may include microbial DNA, precipitation was carried out using 1/10 volume of 3 M sodium acetate (pH 5.2) followed by 0.7 volumes of cold isopropanol. Sodium ions neutralize the negatively charged DNA backbone, while isopropanol reduces the dielectric constant of the solution, decreasing DNA solubility and promoting rapid aggregation and precipitation without overnight incubation.

For termite head-only samples, where cleaner extracts were desired, 3 M potassium acetate (pH 5.5) was used prior to alcohol precipitation. Potassium acetate effectively precipitates SDS–protein complexes, improving DNA purity. The clarified supernatant was then precipitated using either isopropanol or cold ethanol to obtain high-quality genomic DNA.

The DNA pellet was washed with 70% ethanol to remove salts and solvent residues, air-dried briefly, and resuspended in nuclease-free water for immediate enzymatic applications or TE buffer for long-term storage, as EDTA in TE chelates divalent cations and protects DNA from nuclease degradation.

The protocol optimization reduced the duration for DNA extraction to a single day with high yield and quality. The methodology is also informed by species-specific biochemistry through modification (e.g. for termites with potassium acetate, mosquitoes with sodium acetate) of detergents, precipitation agents and buffer systems, at least one of the reasons why this method is particularly effective. Fast isopropanol precipitation and careful use of TE versus NFW make its PCR and sequencing friendly.

3. Comparative Genomic Material Extraction and Quality Assessment

This chapter focuses on both yield and integrity of the extracted DNA. Quality assessment was carried out using nanodrop measurements for purity and concentration, alongside gel electrophoresis to check DNA size and degradation. The methods were optimized for faster processing without compromising results.

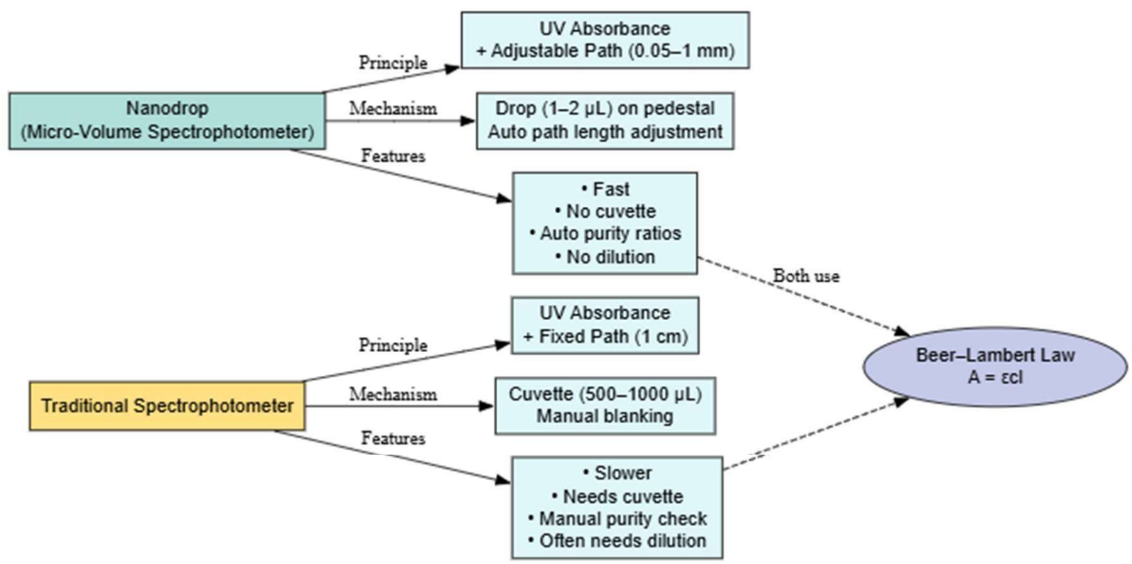


Figure 9: Qualitative Comparison between Quantitative Techniques for Measuring DNA Concentrations.

Figure 10: Nanodrop Readings of extracted DNA

Instrumental differences between NanoDrop & Conventional Spectrophotometer

Nanodrop requires only 1–2 µL samples on a pedestal, and the path length varies to afford rapid, cuvette-less reading to manually measuring absorption at a given wavelength, so that without ever entering in a bias, automatically purity can be derived. The traditional spectrophotometer ought to be used since fixed-path cuvettes had to be used (~1 cm), larger volumes were needed, manual blanking was required, external purity checks were the norm, reading time was longer, although part of this was due to tradeoff between external and internal cell types. Both are based on the Beer–Lambert Law, nonetheless though Nanodrop is faster and is more suitable for molecular work.

Interpretation of absorbance readings and nucleic acid quantification values

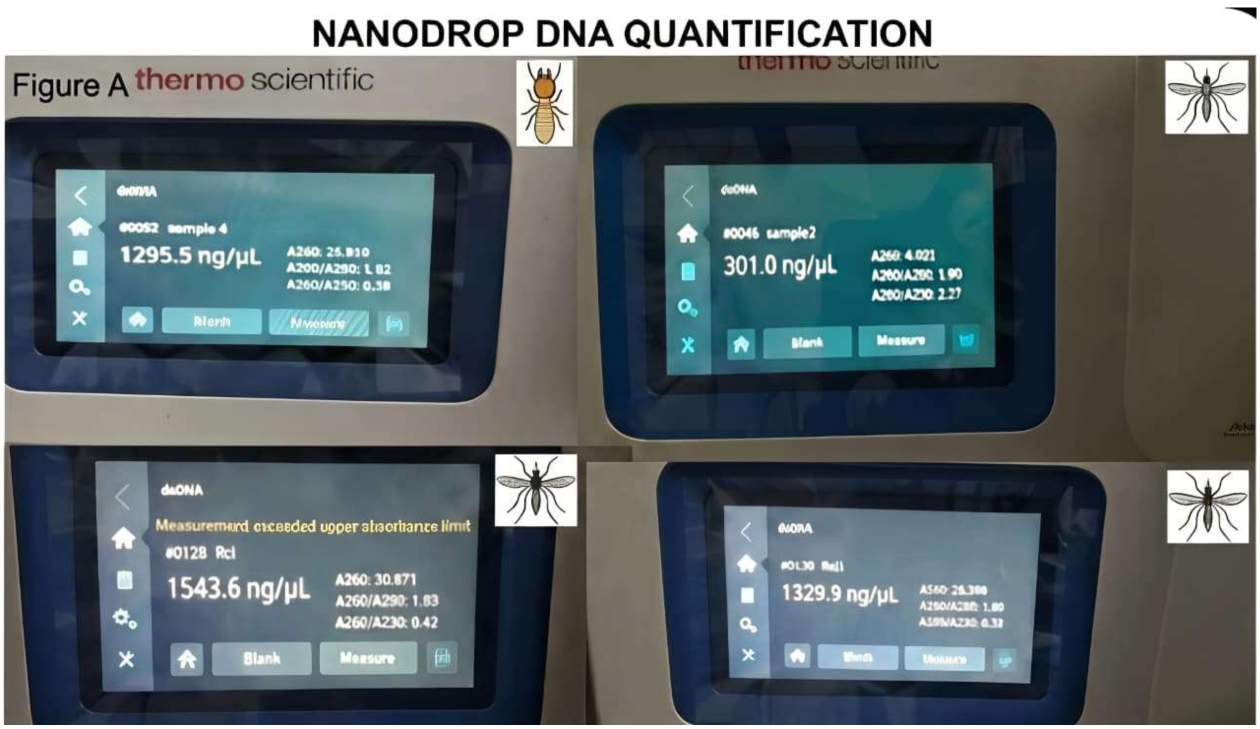


Figure 10: DNA Quantification

Analysis of DNA quality using gel electrophoresis:

Gel Electrophoresis separates nucleic acid fragments by size in an agarose matrix under an electric field. DNA is negatively charged and is drawn toward the positive electrode, with smaller fragments moving more rapidly.

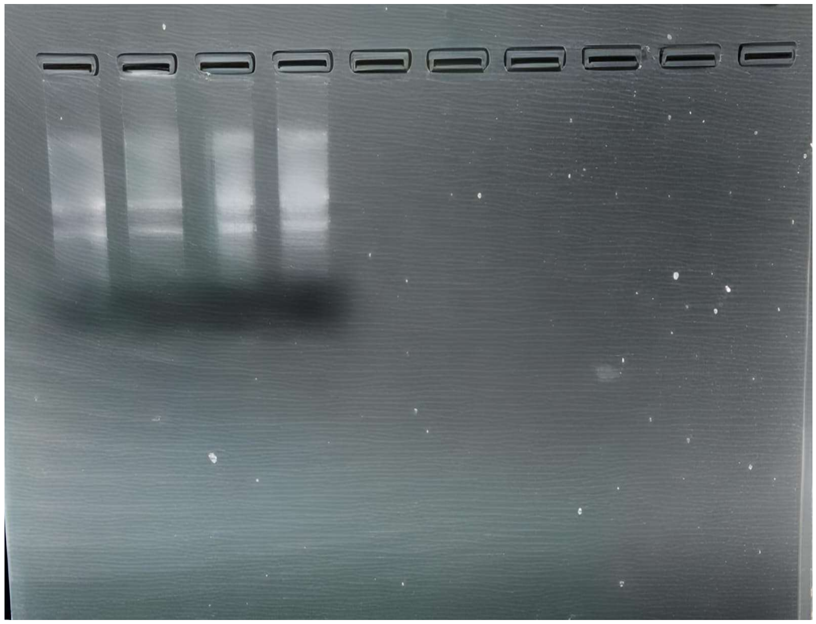


Figure 11: Visible double bands and shearing on single vertical lane

Why 45 minutes and 70 V?

These are standard conditions for resolving fragments of 100 bp to several kb in length. 70 V for a slow even migration and no overheating. ~45 min provides separation of bands without smearing. Higher voltage = faster run, but less resolution and more heat.

Gel Doc (Documentation System):

After electrophoresis, gels stained with ethidium bromide (Intercalating agent) or safer substitutes (e.g., SYBR Safe) are analyzed under UV or blue light with a Gel Doc. It can record the images of DNA bands for documentation and quantification. As two clear bands were visible with a little smear, it is very likely that pattern is consistent → meaning RNA could be present there. Also, the degradation smear is mild → not random contamination or poor extraction, but rather undigested RNA consistently.

4. Functional Biomolecular Analysis and Toxicodynamic Analysis.

Environmental toxicology requires models that respond within the context of their environment organisms that not only absorb contaminants but also show clear, traceable biological responses. Earthworms (Eisenia spp.) have been used in soil studies not because they are easy to handle, but because of their physiology; particularly their oxidative stress machinery, detoxification enzymes, and transcriptional responses which react sharply and reproducibly to xenobiotics in sediment systems. These responses do not float in isolation; they reflect functional impairment.

Danio rerio (Zebra fish), by contrast, allows access to endocrine and developmental processes that are vertebrate specific. With genetic material significantly aligned with humans and transparent embryonic stages, they allow pollutants to be tracked as they interrupt hormone cascades, neural patterning, and organogenesis. These disruptions manifest early and visibly and are supported by transcript-level evidence from well-annotated gene networks.

Figure 12: *Danio rerio* (Zebra fish)

This chapter includes the set of toxicological assays we ran to evaluate oxidative stress and enzyme activity changes after exposure. Tissue sections were taken using a microtome. The goal was to check whether structural changes in organ tissues lined up with what we saw in the biochemical tests. We ran the TBARS assay to measure MDA, which reflects lipid damage under oxidative conditions. GSH levels were checked to get a sense of the antioxidant reserve, and GST activity gave a window into how detoxification pathways were behaving. Protein content was measured using the Bradford method—not just for normalization, but also to see whether stress affected synthesis or degradation. These tests were aimed at mapping whether the exposure caused oxidative pressure or enzymatic disruption at the tissue level. Each of these markers was chosen based on their role in cellular defense or damage response.

4.1. Nuclear Abnormalities and Histological Staining

4.1.1. Micronucleus Test: To detect genotoxic damage, a micronucleus assay was performed on erythrocytes (in zebrafish) and coelomocytes (in earthworms). Smears were prepared on pre-cleaned glass slides, air-dried, then fixed in methanol for 10 minutes. Slides were stained with 5% Giemsa (pH adjusted to ~6.8) for 20–25 minutes. After rinsing gently with distilled water and drying, the slides were observed under a light microscope at 1000x (oil immersion). At least 1000 cells were counted per sample, and the frequency of micronuclei was expressed as a percentage of total cells. Based on representative microscope fields, approximately 140 nucleated cells were observed across selected fields. Among these, 10 cells showed distinct micronuclei. These micronuclei were small, rounded, uniformly stained, and clearly separated from the main nucleus, fulfilling standard scoring criteria. The estimated micronucleus frequency was calculated at roughly 7.14%. This level of nuclear abnormality exceeds typical background frequencies (<0.5–1% in unexposed conditions) and indicates a significant genotoxic effect in the exposed group. Such nuclear damage is consistent with either chromosomal breakage (clastogenic effects) or mitotic spindle disruption (aneugenic effects).

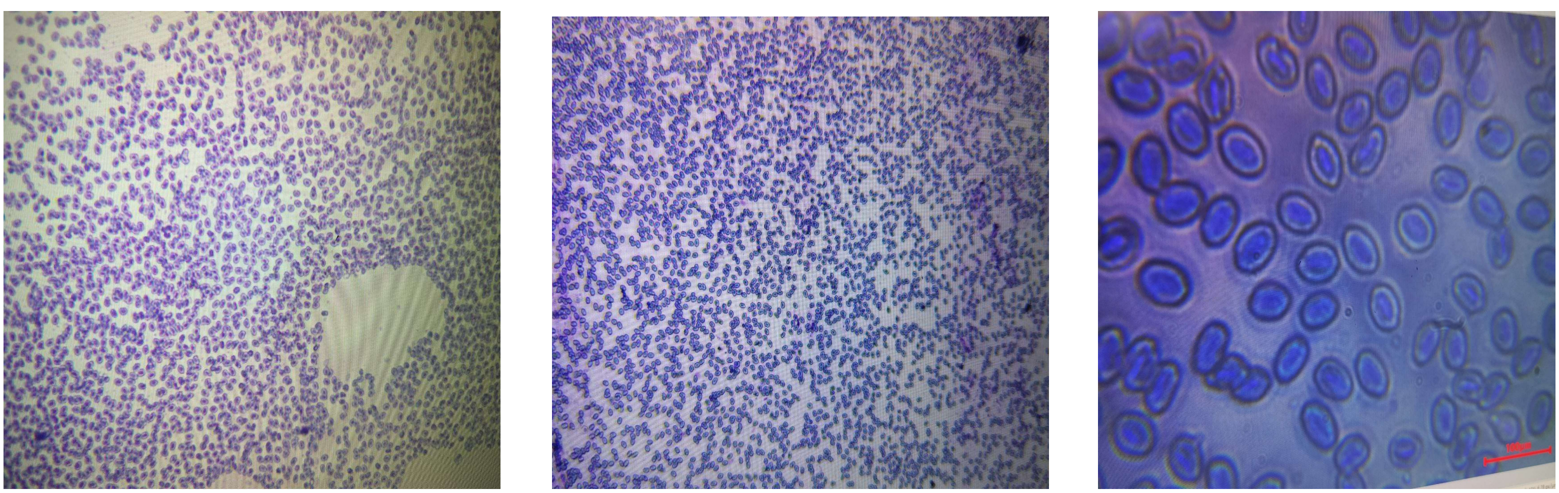


Figure 13: Micronucleus slides: To detect genotoxic damage.

Fig.1. Prepared Wax Blocks of Tissues

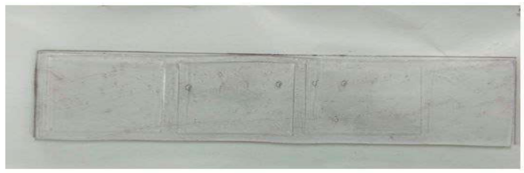


Fig.2. Prepared Histological slide (Eye)

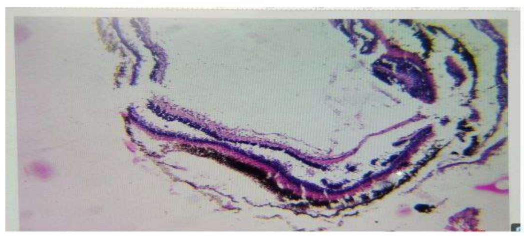


Fig.3.Microscopic view of Slide (Eye)

Figure 14: Preparation and microscopic view of Danio rerio’s eye.

4.1.2. Histological Staining: Fixed tissue samples (liver, gut, eyes) were embedded in paraffin and sectioned at 5 µm thickness using a rotary microtome. Sections were mounted on poly-lysine coated slides and baked at 60°C to enhance adhesion. Routine hematoxylin and eosin (H&E) staining was done to assess structural integrity. Hematoxylin was applied for 6–8 minutes followed by eosin for 30–45 seconds. Sections were dehydrated through graded alcohols, cleared in xylene, and mounted with DPX. Slides were visualized under a compound microscope, and alterations in cellular architecture (e.g., necrosis, vacuolization, epithelial erosion) were noted.

These staining techniques supported the biochemical data, offering cellular and subcellular confirmation of tissue-level toxicity.

4.2. Enzymatic Analysis

Sample Handling and Homogenization

Roughly 100 mg of tissue from either E. fetida (posterior clitellar segments) or pooled adult D. rerio (5–7 per sample) were processed. Tissues were homogenized on ice using 0.1 M phosphate buffer (pH 7.4) containing 1 mM EDTA and 0.1% Triton X-100. The mix was centrifuged at 10,000 rpm for 15 minutes at 4°C. Supernatants were stored at –80°C or used immediately.

MDA Estimation (TBARS Assay)

To assess lipid peroxidation, we quantified malondialdehyde (MDA) via the TBARS method. Each 200 µL aliquot of supernatant was combined with 200 µL of 0.67% thiobarbituric acid (TBA) in 20% acetic acid (pH ~3.5) and 200 µL of 10% trichloroacetic acid (TCA). The samples were heated at 95°C for 30 minutes, cooled, then centrifuged at 4000 rpm for 10 minutes. Absorbance was recorded at 532 nm. A TMP-based standard curve was used to calculate MDA equivalents.

GSH Quantification (DTNB Method)

Reduced glutathione was measured using DTNB, which reacts with thiol groups to form a yellow chromophore detectable at 412 nm. To start, samples were deproteinized with 5% TCA and centrifuged. A portion of the supernatant was then added to freshly made DTNB in phosphate buffer (0.01 M). After a short room-temperature incubation, readings were taken. A standard curve from known GSH concentrations (0 to 100 µM) was plotted alongside. This method gives a rough estimate of antioxidant reserve remaining in the tissue.

GST Activity

GST activity was checked using CDNB and GSH in phosphate buffer. The mixture was assembled quickly and read at 340 nm over several minutes. The rise in absorbance shows conjugation activity. Higher readings usually point toward induced detox pathways, possibly in response to sustained exposure.

Total Protein (Bradford Method)

Protein content was estimated using the Bradford method. Coomassie dye was added to samples, and absorbance was read at 595 nm after 5 minutes. BSA was used for standards. This allowed each oxidative stress marker to be normalized to total protein.

Instrument Settings and Controls

All assays were performed in 1 cm pathlength quartz cuvettes or flat-bottom microplates. Blank wells included all reagents minus the sample. Light-sensitive chemicals like DTNB and CDNB were stored in foil-wrapped tubes. Over-range readings were handled by diluting samples 1:2 and rerunning the assay.

Interpreting Patterns

When MDA went up, and GSH levels dropped, and GST activity increased, this combination typically marked oxidative stress. Looking at how each organism responded under similar exposure helped us catch differences that would be missed with just one species. These biochemical changes often occurred before anything was visible under the microscope.

This workflow kept things functional and replicable without needing high-end instrumentation. It suited both standard lab setups and minimal-resource conditions.

4.3. Haematological Indices and Dilution Factors (Additional informations)

RBC and WBC Dilution Protocols

Red Blood Cells (RBC) were diluted using Hayem's solution or isotonic saline in a 1:200 ratio while White Blood Cells (WBC) were diluted using Turk's solution in a 1:20 ratio. The appropriate diluting fluids preserved cell integrity while lysing non-target elements (for WBCs).

Neubauer Chamber

A Neubauer counting chamber was used for manual blood cell enumeration. After proper dilution and mixing, a drop was loaded into the chamber using a clean pipette. The chamber was left undisturbed for 2–3 minutes to allow cells to settle.

- RBCs were counted in five small squares of the central large square.
- WBCs were counted in the four large corner squares.

Cell count (cells/μL) = (Number of cells counted × dilution factor × 10) / Area counted (mm^2)

Hematocrit Measurement

Hematocrit values were determined using capillary tubes filled with fresh anticoagulated blood. Tubes were centrifuged in a micro-hematocrit centrifuge for 5 minutes at ~12,000 rpm.

- Packed Cell Volume (PCV) was calculated as the ratio of red cell column height to total blood column height.
- Hematocrit results reflected hydration status, oxygen-carrying capacity, and possible hemolytic damage.

Together, hematological parameters offered an additional layer of interpretation, correlating physiological stress with biochemical and nuclear endpoints.

5. Supplementary Experimental Techniques and Analytical Exposure

5.1. INTRODUCTION TO ELISA

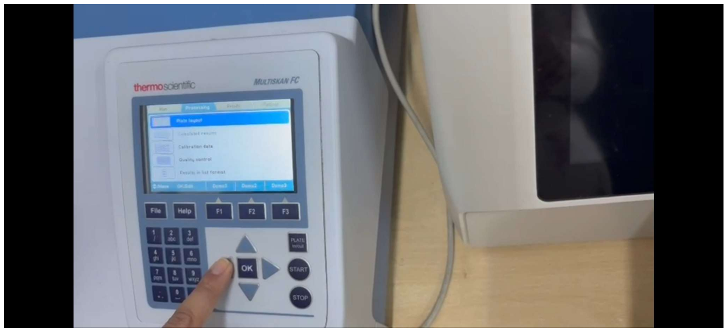
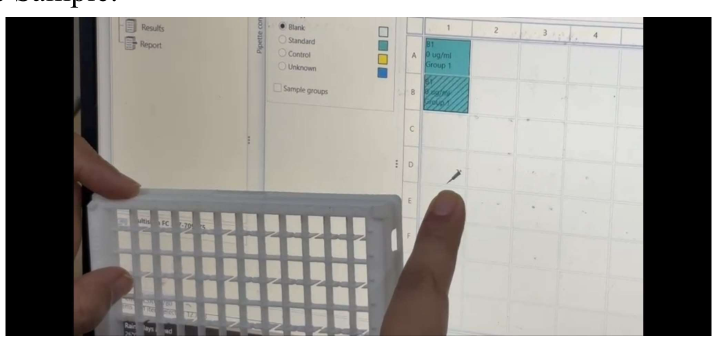


Figure 15: ELISA Operations

The ELISA (Enzyme-Linked Immunosorbent Assay) procedure was demonstrated by Miss Navya. While no experimental work was carried out by students during this session, the tutorial focused on explaining the operational steps and principles behind the technique. The following aspects were covered:

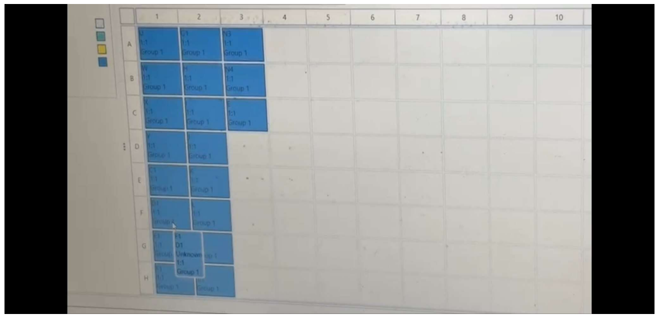
Coating of the wells with antigen or antibody. Blocking to prevent non-specific binding. And addition of primary and secondary antibodies, including enzyme-linked antibodies. Substrate reaction leading to a measurable color change. Interpretation of results using absorbance readings. .

15 a) Loading the Sample:



First, a 96-well plate was used. Each well was filled with a sample using a micropipette. Care was taken to avoid bubbles and pipette correctly into the center of each well. The plate was then covered and kept for incubation. After incubation, the plate was washed using a plate washer or manually with buffer solution. This step removes any unbound substances. Enzyme-linked antibodies and a substrate solution were added. A color change occurred if the target substance was present.

15 b) Operating from Laptop:



The plate was placed into the ELISA reader machine. The machine was connected to a laptop via USB or Bluetooth. A software (like MicroWin, SoftMax Pro, or the software specific to the machine) was opened. The program was set to measure absorbance (usually at 450 nm). After running, the results were shown on the screen in the form of absorbance values for each well.

15 c) Interpreting Results:

The software displayed a table or graph. Higher color intensity (higher absorbance) meant a stronger positive result.

5.2. General Procedures:

All analytical procedures were conducted under standard laboratory conditions. Glassware was acid-washed and rinsed with deionized water. Reagents were prepared fresh or sourced from validated commercial stocks, ensuring consistency across assays. Instruments were routinely calibrated. Sample handling followed cold chain protocols when necessary.

5.2.1. Total Dissolved Solids (TDS)

TDS was assessed using a conductivity-based meter equipped with a temperature-compensated probe. Calibration involved a 1000 ppm NaCl standard. Probes were cleaned with distilled water and stored dry between uses. Samples were briefly agitated before measurement to minimize stratification. Results were expressed in mg/L.

5.2.2. pH Determination

A digital benchtop pH meter, calibrated using NIST-traceable buffer solutions at pH 4.0, 7.0, and 10.0, was employed. Electrodes were rinsed between samples with distilled water and gently blotted dry. Measurements were taken after stabilization of the reading. Buffer solutions were replaced weekly or sooner if contamination was suspected.

5.2.3. Blood Pressure, Glucose and Haemoglobin: As part of the supplementary physiological assessment, three non-invasive or minimally invasive metrics blood pressure, blood glucose, and haemoglobin concentration—were recorded to provide supporting evidence of systemic stress or early physiological alteration. While not part of the primary toxicological endpoints, these parameters offered ancillary insight into organismal response under exposure.

Blood pressure measurements were carried out using a digital sphygmomanometer . Subjects were positioned at rest, and readings were taken from the upper arm using the auscultatory method. As the cuff deflated, the device detected oscillations in arterial wall movement, from which systolic and diastolic pressures were algorithmically derived. These measurements were used to assess changes in cardiovascular tone, potentially associated with fluid imbalance, autonomic stress, or endocrinerelated vascular effects.

Glucose levels were assessed via fingertip capillary blood using a glucometer based on the glucose oxidase-peroxidase reaction. Glucose fluctuation served as an indirect marker for metabolic stress or compensatory endocrine responses especially relevant where energy regulation may be impaired by pollutant-induced hormonal disruption.

Haemoglobin concentration was estimated from venous blood samples collected in EDTA-treated tubes. The cyanmethemoglobin method was used for quantitative determination, with absorbance measured at 540 nm following stabilization in Drabkin's reagent. A portable hemoglobinometer was employed where field constraints limited lab-based processing. Hemoglobin values were examined to identify anemia or impaired oxygen transport, which may arise from chronic exposure, oxidative stress, or hematopoietic suppression.

Together, these supplementary parameters contributed to a broader physiological context for interpreting observed effects. While not diagnostic on their own, they provided additional granularity when evaluating the biological relevance of primary findings.

5.3. Digital Softwares

To support analytical clarity and evidence mapping, three digital tools were explored. Graphviz enabled the construction of mechanistic diagrams using DOT language, visually linking pollutants to physiological outcomes. VOSviewer was used for bibliometric clustering, revealing thematic trends and research gaps from literature databases. Scite.ai aided citation triage by distinguishing supporting, contrasting, and neutral references, streamlining literature curation for mechanistic validation.